5. Online Payment System

Program :

```
from abc import ABC, abstractmethod

# Abstraction
class Payment(ABC):

    def __init__(self, amount):
        self.amount = amount

    @abstractmethod
    def pay(self):
        pass

# Inheritance
class UPI(Payment):

    def __init__(self, amount):
        super().__init__(amount)

    # Polymorphism
    def pay(self):
        print(f"Paid ₹{self.amount} using UPI")

class Card(Payment):
```

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```
    def __init__(self, amount):
        super().__init__(amount)

    # Polymorphism
    def pay(self):
        print(f"Paid ₹{self.amount} using Card")

class Cash(Payment):

    def __init__(self, amount):
        super().__init__(amount)

    # Polymorphism
    def pay(self):
        print(f"Paid ₹{self.amount} using Cash")

# ---- Main Program ----

mode = input("Enter Payment Mode (UPI/Card/Cash): ")
amount = float(input("Enter Amount: "))

if mode.lower() == "upi":
    payment = UPI(amount)
elif mode.lower() == "card":
    payment = Card(amount)
elif mode.lower() == "cash":
    payment = Cash(amount)
else:
```

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```
    print("Invalid Payment Mode")
    exit()

payment.pay()
```

Output :

Enter Payment Mode (UPI/Card/Cash): UPI

Enter Amount: 100

Paid ₹100.0 using UPI

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